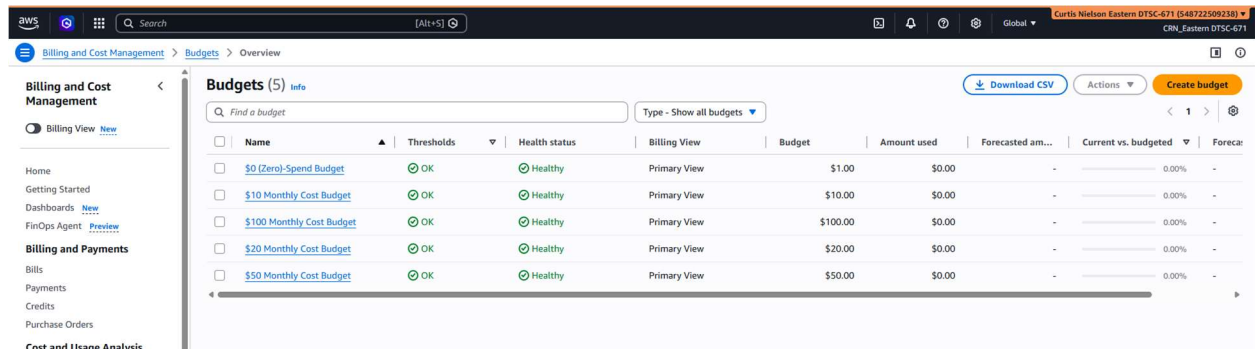


# DTSC 671: AI Solutions in the Cloud

## Assignment: Budget and Pricing Calculator

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### Part 1

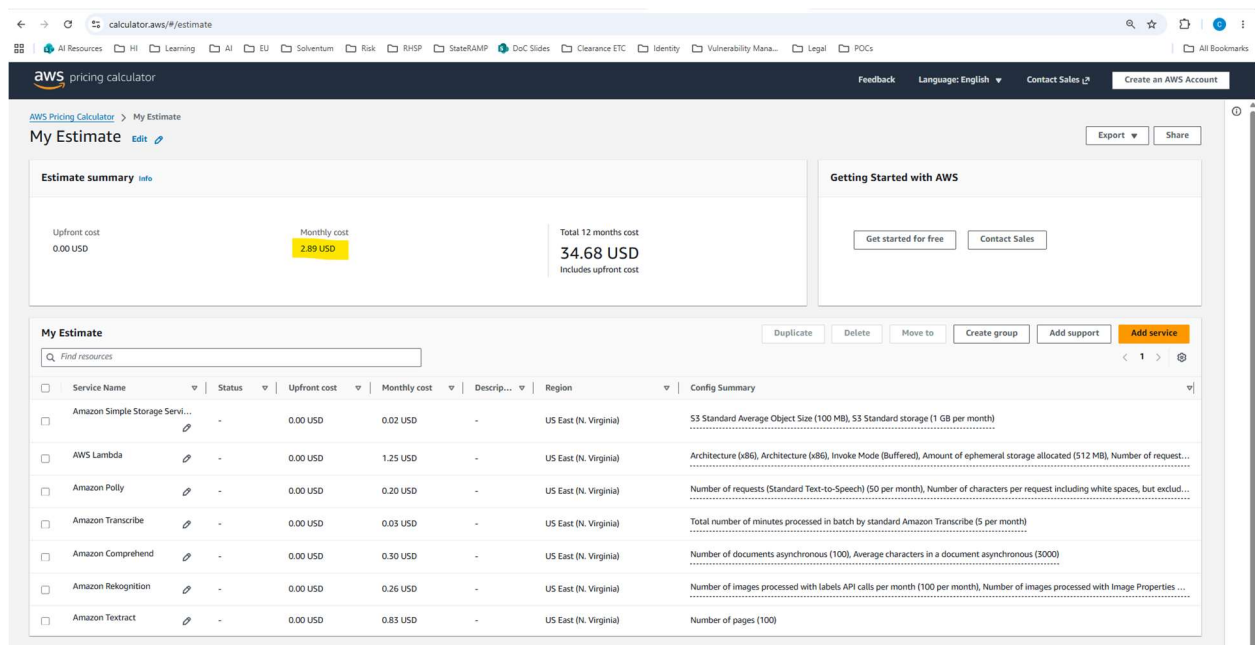


The screenshot shows the AWS Billing and Cost Management console. On the left is a navigation menu with options like Home, Getting Started, Dashboards, FinOps Agent, Billing and Payments, and Cost and Usage Analysis. The main area is titled 'Budgets (5)' and contains a table of five budgets. Each budget has a name, thresholds, health status, billing view, budget amount, amount used, forecasted amount, and current vs. budgeted percentage. All budgets are in 'Primary View' and have a 'Healthy' status.

| Name                                      | Thresholds | Health status | Billing View | Budget   | Amount used | Forecasted am... | Current vs. budgeted | Foreca |
|-------------------------------------------|------------|---------------|--------------|----------|-------------|------------------|----------------------|--------|
| <a href="#">\$0 (Zero)-Spend Budget</a>   | OK         | Healthy       | Primary View | \$1.00   | \$0.00      | -                | 0.00%                | -      |
| <a href="#">\$10 Monthly Cost Budget</a>  | OK         | Healthy       | Primary View | \$10.00  | \$0.00      | -                | 0.00%                | -      |
| <a href="#">\$100 Monthly Cost Budget</a> | OK         | Healthy       | Primary View | \$100.00 | \$0.00      | -                | 0.00%                | -      |
| <a href="#">\$20 Monthly Cost Budget</a>  | OK         | Healthy       | Primary View | \$20.00  | \$0.00      | -                | 0.00%                | -      |
| <a href="#">\$50 Monthly Cost Budget</a>  | OK         | Healthy       | Primary View | \$50.00  | \$0.00      | -                | 0.00%                | -      |

Figure 1: 5 Budgets in personal AWS account

### Part 2



The screenshot shows the AWS Pricing Calculator interface. It displays an estimate for various services. The 'Estimate summary' section shows an upfront cost of 0.00 USD and a monthly cost of 2.89 USD, totaling 34.68 USD for 12 months. The 'My Estimate' table lists services like Amazon S3, AWS Lambda, Amazon Polly, Amazon Transcribe, Amazon Comprehend, Amazon Rekognition, and Amazon Textract, along with their respective costs and configurations.

| Service Name                   | Status | Upfront cost | Monthly cost | Descrip... | Region                | Config Summary                                                                                                                       |
|--------------------------------|--------|--------------|--------------|------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Amazon Simple Storage Servi... | -      | 0.00 USD     | 0.02 USD     | -          | US East (N. Virginia) | S3 Standard Average Object Size (100 MB), S3 Standard storage (1 GB per month)                                                       |
| AWS Lambda                     | -      | 0.00 USD     | 1.25 USD     | -          | US East (N. Virginia) | Architecture (x86), Architecture (x86), Invoke Mode (Buffered), Amount of ephemeral storage allocated (512 MB), Number of request... |
| Amazon Polly                   | -      | 0.00 USD     | 0.20 USD     | -          | US East (N. Virginia) | Number of requests (Standard Text-to-Speech) (50 per month), Number of characters per request including white spaces, but exclud...  |
| Amazon Transcribe              | -      | 0.00 USD     | 0.03 USD     | -          | US East (N. Virginia) | Total number of minutes processed in batch by standard Amazon Transcribe (5 per month)                                               |
| Amazon Comprehend              | -      | 0.00 USD     | 0.30 USD     | -          | US East (N. Virginia) | Number of documents asynchronous (100), Average characters in a document asynchronous (5000)                                         |
| Amazon Rekognition             | -      | 0.00 USD     | 0.26 USD     | -          | US East (N. Virginia) | Number of images processed with labels API calls per month (100 per month), Number of images processed with Image Properties ...     |
| Amazon Textract                | -      | 0.00 USD     | 0.83 USD     | -          | US East (N. Virginia) | Number of pages (100)                                                                                                                |

Figure 2: Estimate 1 Screenshot (S3, Lambda, Polly, Transcribe, Comprehend, Rekognition, and Textract)

**Question #1:** Based on these calculations, what is an estimate for completing the labs and assignments in this course?

**Answer:** The course is less than 2 months long, and the estimated monthly cost (highlighted) is 2.89 USD, so the estimated cost for completing the labs and assignments is less than  $2 \times 2.89$  USD or **< 5.78 USD**.

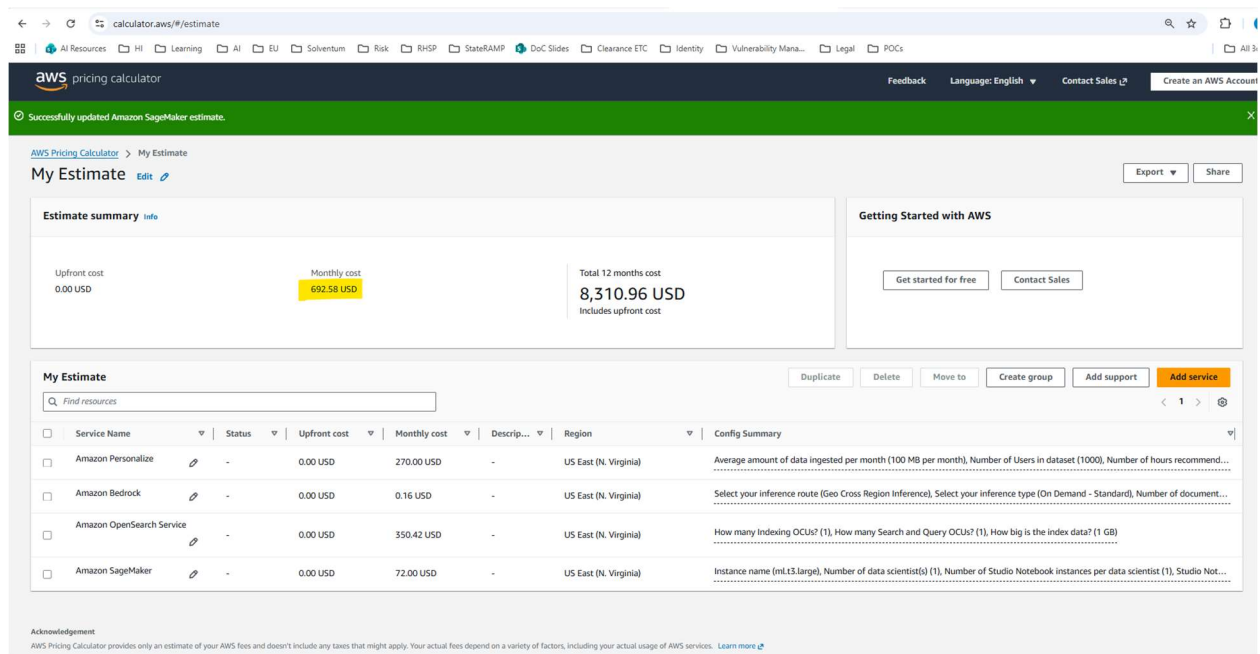


Figure 3: Estimate 2 Screenshot (Personalize, BedRock, OpenSearch Serverless, and SageMaker)

**Question #2:** Based on these calculations, what is an estimate for your monthly bill if you're not careful with these services?

**Answer:** The estimated monthly bill if the services above are left running is **692.58 USD** (highlighted).